Module 3 Challenge

Calculus Honors · Continuity & the Great Theorems · 16 points

Name: _____ Date: _____

> This is your fullest, most careful work of the module. Show every step in clean vertical layout, justify your reasoning in words where asked, and label your final answers. You will be graded on understanding, quality of reasoning, clarity and precision, and intellectual honesty (say clearly where you are certain and where you are not).

1. Continuity at a seam (4 pts). Let $f(x) = \begin{cases} x^2 - 1 & x < 2 \\ 3x - 5 & x \geq 2 \end{cases}$. Test whether f is continuous at $x = 2$ by checking all three conditions in order, showing each one explicitly. State your conclusion, including which condition, if any, fails.

2. Classify every discontinuity (4 pts). Let $h(x) = \dfrac{x - 1}{(x - 1)(x + 3)}$. Find every value of x where h is discontinuous, and classify each discontinuity as removable, jump, or infinite. Justify each classification using one-sided limits.

3. Prove a root exists (4 pts). Prove that $g(x) = x^4 - 3x - 2$ has a root on the interval $[1, 2]$. State all four steps of the argument: continuity, endpoint values, sign change, and conclusion, citing the Intermediate Value Theorem by name.

4. Check the hypotheses, and build a counterexample (4 pts). A function k is continuous on the open interval $(-2, 5)$. Does the Extreme Value Theorem guarantee that k attains an absolute maximum on this interval? Explain which hypothesis is or is not met. Then construct a specific example of a continuous function on $(-2, 5)$ that has no absolute maximum, and briefly explain why your example works.